

Augmentation of Medical Preparation for Children by Using Projective and Tangible Interface

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Figure 1: (A)Scene of the use(B)Scene of the deformed character projected onto the soft toy's body(C)Hospital Toy Devices. (C-1)Syringe-type (C-2)Stethoscope-type (C-3)Thermometer-type

ABSTRACT

This research aims to create an interactive experience that alleviate anxiety of pediatric patients and cause empathy within their family and medical community. We developed the novel medical preparation system through the integration of projective and tangible interfaces. Through our work, children can intuitively understand their illnesses and medical procedures in an age-appropriate and engaging manner.

CCS CONCEPTS

• **Human-centered computing** → Mixed/augmented reality; Virtual reality; • **Applied computing** → Interactive learning environments.

KEYWORDS

Children, Interactive art, Augmented reality, Assistive technology, Pediatric care, medical play, patient preparation, Alleviate anxiety, Hospitalized children's well-being, Child-centered design

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1 INTRODUCTION

In pediatric care, it is important to protect the rights of children in hospitals such as the Convention on the Rights of the Child adopted by the United Nations [United Nations, 1989] and the Charter for Children in Hospitals established by the European Association for Children in Hospital [European Association for Children in Hospital, 1993]. Based on these children's rights, it is necessary to provide medical play and education including preparation to promote the healthy development of children's minds and bodies and also to alleviate the stress and anxiety in pediatric care [Godino-láñez et al., 2020] [Koukourikos et al., 2015].

Several studies have been conducted using interactive technology to alleviate stress of children through play in hospitals. Jeong et al. studied how virtual agents of social robots affect pediatric patients' emotions, fun play, and social interactions with others [Jeong et al., 2018]. Liszio et. al. investigated playful MRI exposure experience in a virtual world reduces children's anxiety and stress about MRI examinations [Liszio and Masuch, 2017]. Sonne et al. developed breath-controlled biofeedback game technology with bespoke airflow sensor that aims to calm children during blood sampling procedures [Sonne et al., 2017].

In our approach, we utilize the characteristics of interactive art to facilitate their storytelling that helps children overcome illness through the artwork. Furthermore, our artwork fosters connections between children and their families with other children who are fighting illnesses similar to their own and encourages them to connect with medical communities and facilities (Figure 1).

2 IMPLEMENTATION

Respecting the rights of children in difficulties of illness, our artwork aims to bring their inherent resilience and foster a sense of resonance within their community. In this work, children can intuitively understand what their illnesses are and what kind of

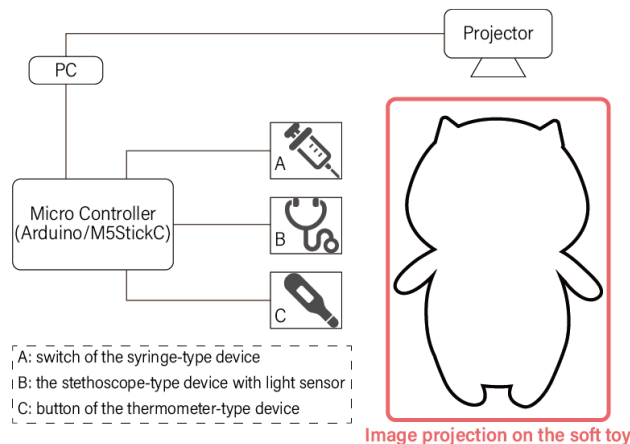


Figure 2: System configuration

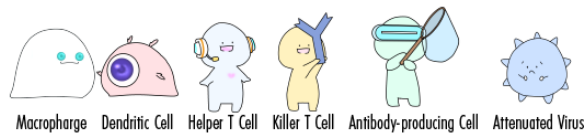


Figure 3: The character of medical elements

treatment they need by using our Hospital Toy Devices, which mimics real medical equipment, to administer treatment to the soft toy in their play. It enables children to engage in medical play, fostering trust in medical care as they share these experiences with healthcare professionals in positive feeling. Additionally, it provides opportunities for children to interact with others facing similar illnesses, promoting mutual understanding, and empathizing loneliness and pain. Moreover, this artwork facilitates conversations and closer relationships with their families may in communication gaps due to hospitalization.

In our prototype, we developed three hospital toy devices such as a syringe-type toy, stethoscope-type toy and a thermometer-type toy. When children play with the toy device, images are projected onto the soft toy. For Example, the piston of the syringe-type toy device is pressed, an animation illustrating the mechanism of the vaccine is projected onto the surface of the soft toy's body. The stethoscope-type toy device, when placed against the soft toy, enhances the animation's sound volume through an incorporated light sensor, enabling children to hear the changes occurring within the soft toy's body. The thermometer-type device alters the displayed temperature value when its button is pressed. This change is reflected in the soft toy's facial color, which corresponds to the temperature (Figure 2).

The soft toy functions as a screen for images to explain the treatment and its necessity. For example, the body of soft toy will be projected the animated representations of viruses, cells, and other medical elements to make it easily understandable and relatable for children and incorporating sound effects to bring their imagination (Figure 3).

3 CONSIDERATION

As a preliminary step to conducting experiments with children, verification experiments were conducted with 56 adults. In response to the question of whether our artwork enables children who struggle with illnesses or injuries to deepen their understanding of the treatment, 98.2% of the participants answered, "strongly agree" or "agree". Reasons cited included "feeling familiar with cute characters and the soft toy" (37.5%) and "reducing fear of the unknown" (18.8%). Moreover, responses suggested that it is effective in alleviating children's stress, attributed to reasons such as "being visually approachable for understanding" (34.4%) and "clear understanding through experiential explanations" (15.6%).

According to the above results, it is considered that beyond reducing children's fear of the unknown, such as illness and treatment, the modeling of animated characters and the inherent sense of security in soft toys is also effective.

In addition, some positive comments were received such that children and families can experience together our artwork and get the opportunity to have conversations about the disease and its treatment in a pleasant atmosphere.

4 CONCLUSION

Our artwork including Hospital toy devices, resembling various medical equipment, not only function as triggers for visuals and sounds, but also encourage an interest in medical play. Furthermore, our artwork has a high potential to be used as a preparation device to explain the content and significance of treatments and procedures. This could also contribute to stress relief among children. Currently, we focus on vaccinations, but our artwork can be adapted to other diseases. We are preparing to improve our work in facilities such as children's hospitals with the cooperation of experts who are doctors and specialists in childcare such as HPS and CLS.

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